CS & IT ENGINEERING



Compiler Design

12 Problems

Parser

DPP – 02 Discussion Notes



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[MCQ]

#Q. Consider the following CFG:

$S \rightarrow 1SP2 \mid 2SQ3$

$P \rightarrow 5AB \mid \varepsilon$

$Q \rightarrow 6BC \mid \varepsilon$

$R \rightarrow 1C \mid 4$

In above grammar S is a start variable, P, Q and R are non terminals, {1, 2, 3, 4, 5, 6} are terminals. What is in the Follow(S)?

Follow(S) = First(P2) U First(Q3)

= {5, 2} U {6, 3}

= {2, 3, 5, 6, $}

A {1, 2, 3, 5, $}

B {1, 3, 5, 6, $}

C {2, 3, 6, 5, $}

D {1, 2, 4, 6, $}

[MCQ]

#Q. Consider the following CFG:

S → aSAb | bSBc

A → A+aB | ε

A → B*bC | ε

B → aC | d | ε

What is in the First(A)?

First(A) = {ε, +, a, d, *}

First(B) = {a, d, ε}

First(B*bC) = {a, d, ε, *}

A {+, ε, a, d, *}

B {ε}

C {ε, +, a, d}

D {+, ε}

[MSQ]

#Q. Which of the following is/are CORRECT statement?

A, C, D

A. Every left factored grammar can not be LL(1). True

B. The class of grammars that can be parsed using LR methods is proper subset of the class of grammar that can be parsed by LL method

C. LR parsing is non-backtracking method

↳ Because we use Parse Table

D. If grammar is LL(1) then it will be LR(1) also.

① Left | LF | Amb ↳ Not LL(1), ↳ CLR

LR1

↓

Not LR(0), SLR, CLR, LALR

LR ⊂ LL

(LL ⊂ (LR)

LL(1) → CLR(1)

[MSQ]

#Q. Consider the following two grammars

G1: S → S1 | 0S1 | 1

G2: P → P0| 1

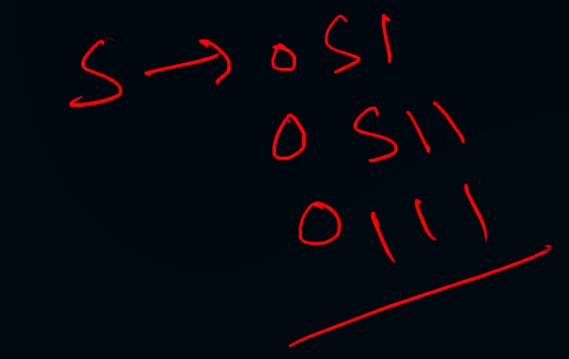
Which of the following is True regarding above grammars?

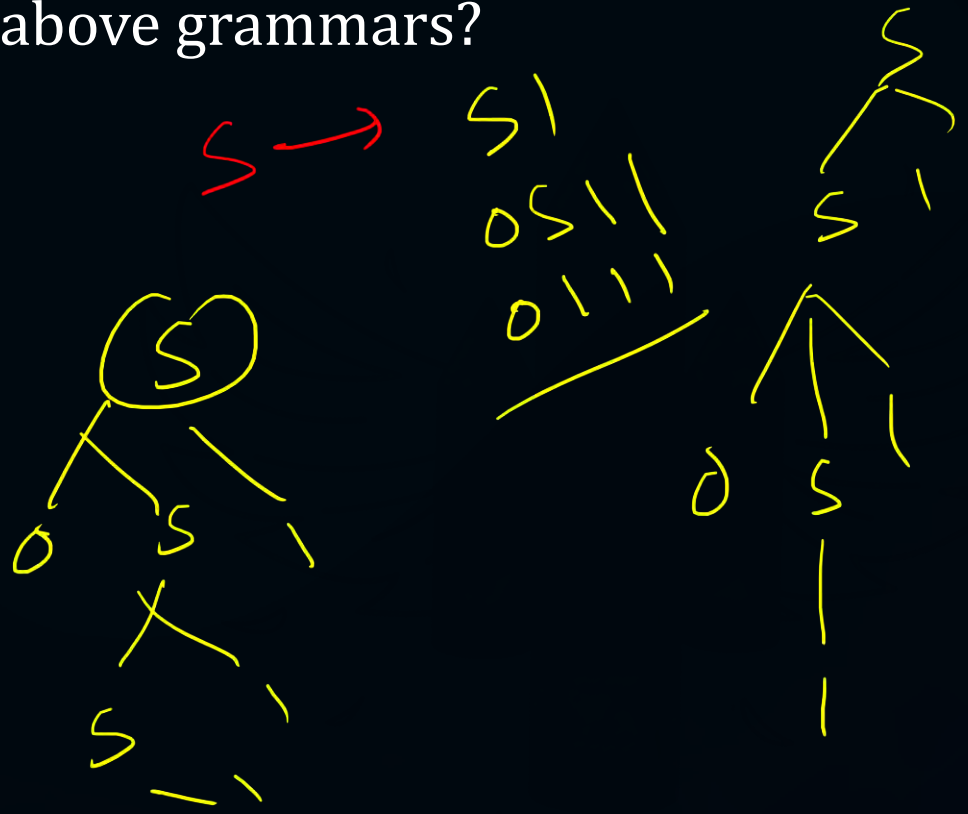
A. G_1 is not LR(1)

B. G_2 is LL(1)

C. Both G_1 and G_2 is LL(1)

D. G_2 is not LL(1)





[NAT]

#Q. Consider the following grammar.

$S \rightarrow aB \mid aAb$

$A \rightarrow bAb \mid a$

$B \rightarrow aB \mid \varepsilon$

How many parse trees are possible to generate the string "aab" from the above grammar? 1





[NAT]

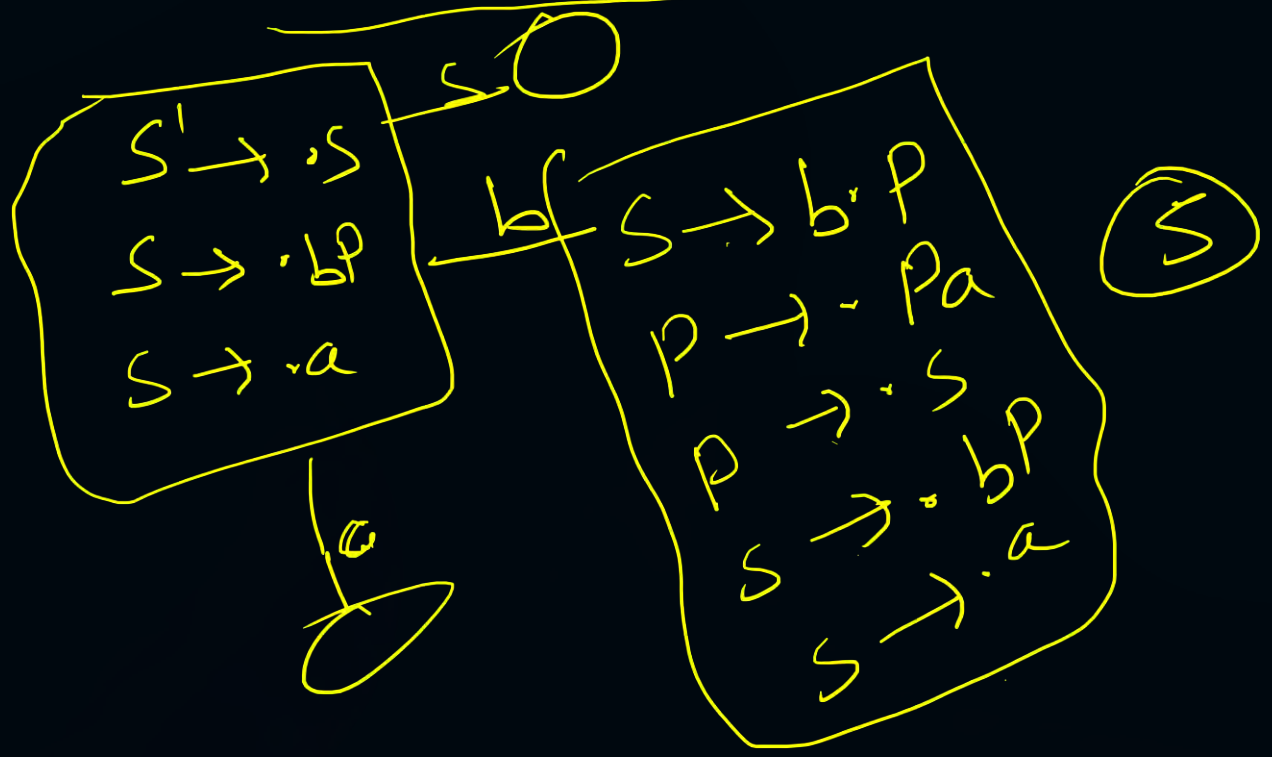
#Q. Consider the augmented grammar given below

S'→S

S→ bP | a

P→ Pa | S

Let I_0 = CLOSURE({[S'→.S]}). The number of items in the set GOTO(I_0,b) is:_____



[NAT]

#Q. Consider the following grammar.

$S \rightarrow aSB \mid c$

$B \rightarrow b$

The number of reduction steps taken by a bottom-up parser while accepting the string "aaacbbb" is _____.



[NAT]

#Q. Consider the following statements:

I. Canonical LR is more powerful than LL parser. → True

II. Every SLR(1) grammar is also an LALR(1) grammar. → True

III. Every LL(1) grammar is also an CLR(1) grammar. → True

The number of incorrect statements is/are ___0___.



[MCQ]

#Q. Consider the given grammar G:

$A \rightarrow aA \mid Aab \mid b$

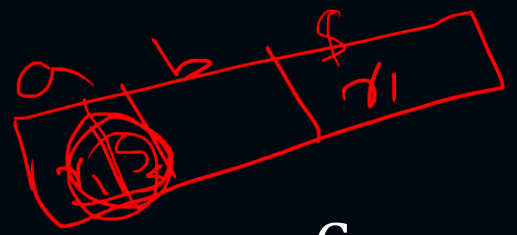
Select the correct option.

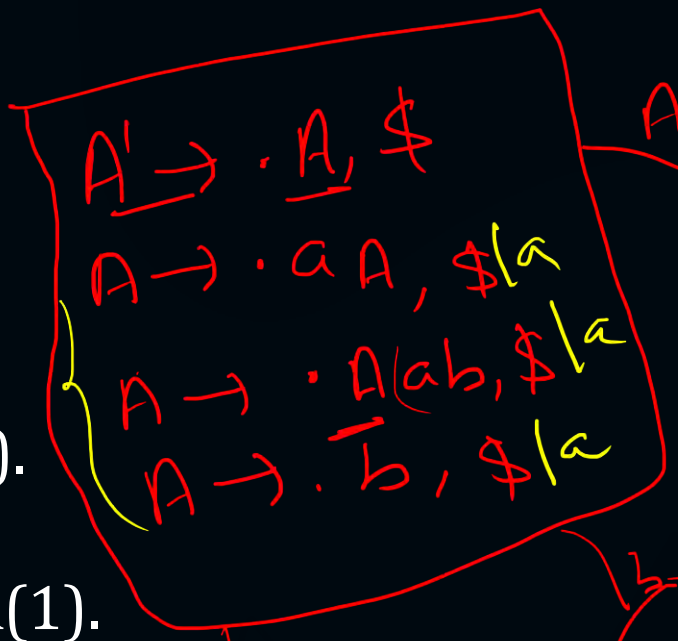
A. Given grammar is SLR(1) but not LR(0).

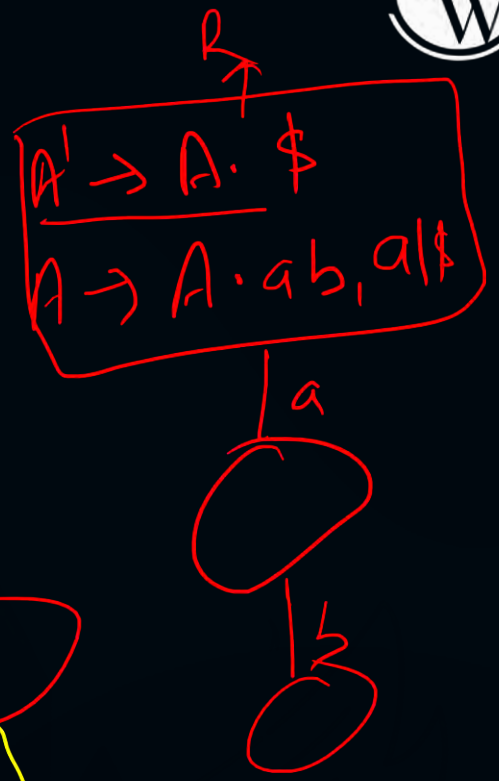
B. Given grammar is LALR(1) but not SLR(1).

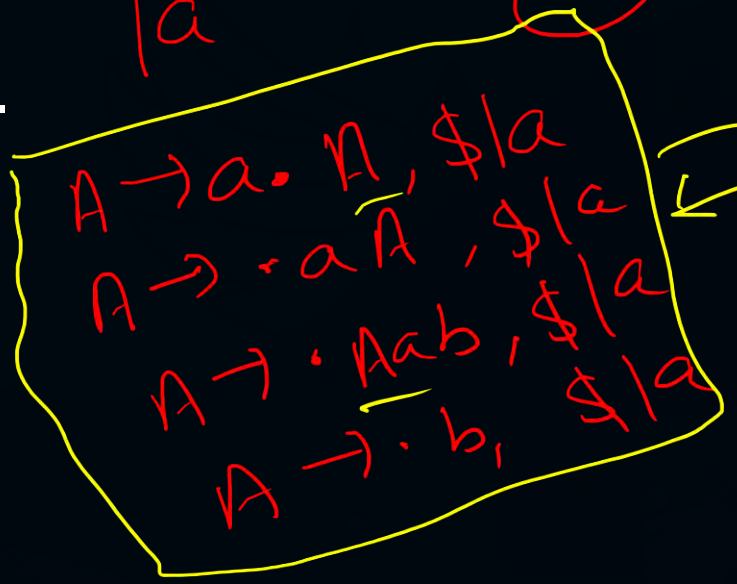
C. Given grammar is CLR(1) but not LALR(1).

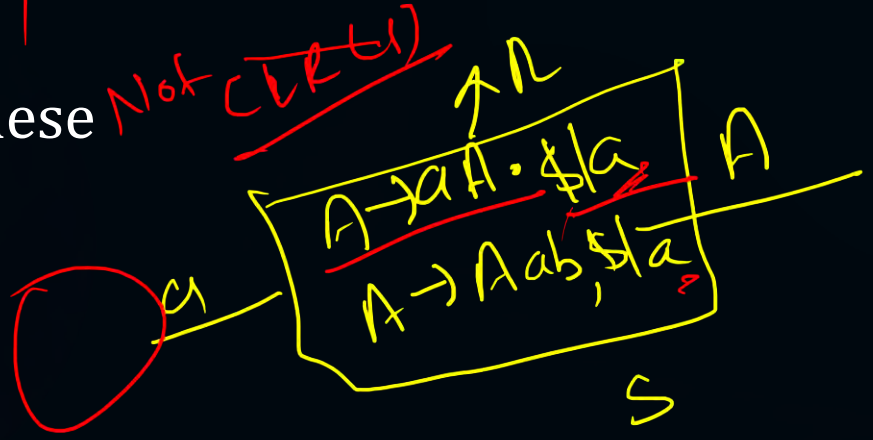
D. None of these











[MCQ]

#Q. Consider the given below statements.

S_1: Every CLR(1) grammar is unambiguous.

S_2: Every unambiguous grammar is CLR(1)

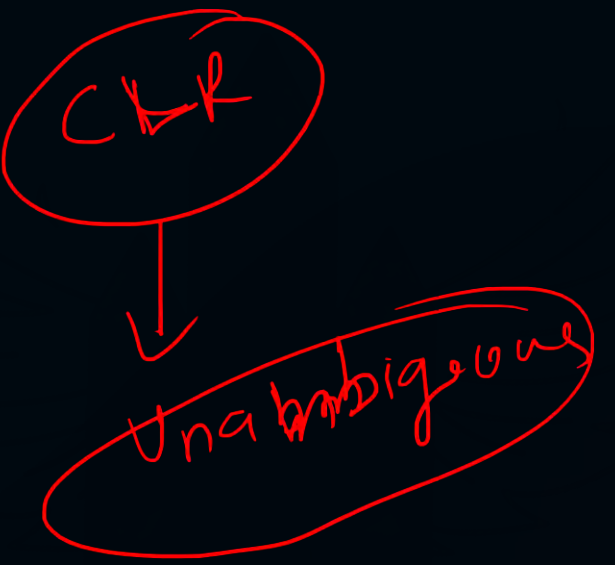
Select the correct statement:

A. Both S_1 and S_2 are correct.

B. Both S_1 and S_2 are incorrect.

C. S_1 is correct while S_2 is incorrect.

D. S_1 is incorrect while S_2 is correct.



[MCQ]

#Q. Consider the following statements

S1: If a context free grammar is unambiguous then the language generated by this grammar must be DCFL.

S2: Every CFG corresponding to a DCFL must be unambiguous.

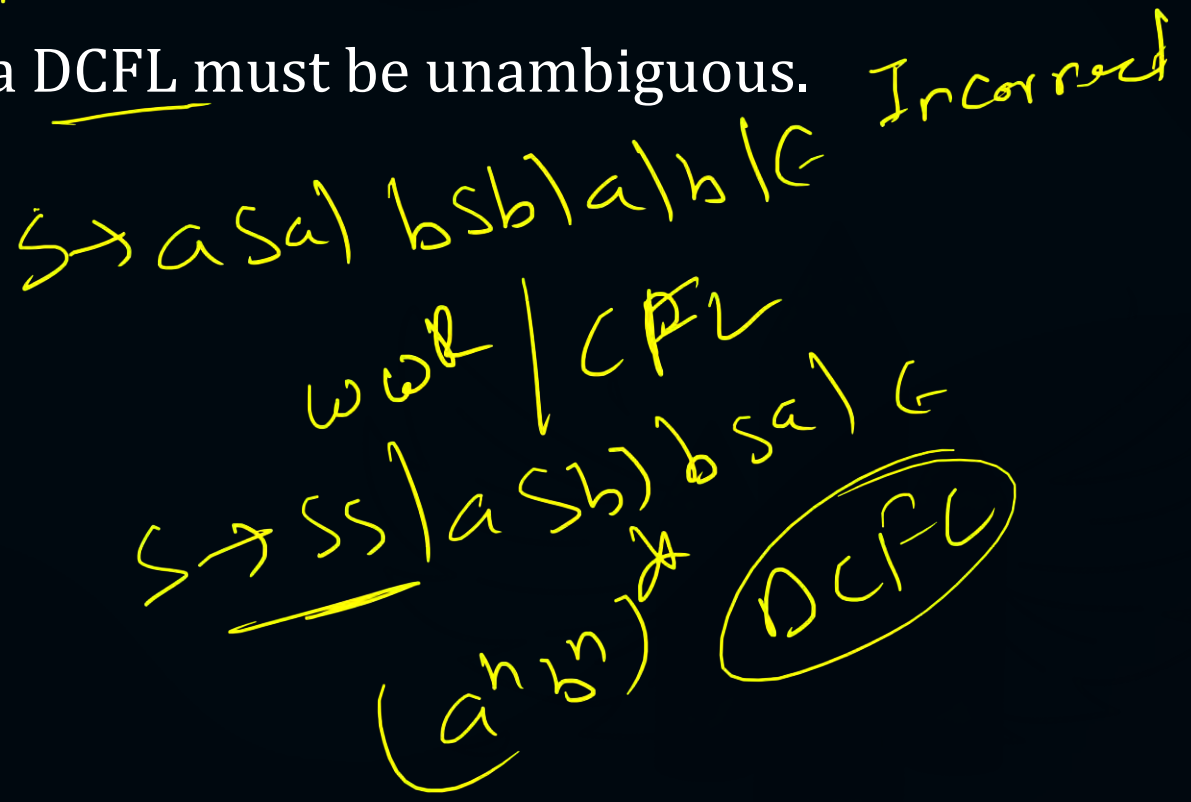
Select the correct option.

A Both S1 and S2 are true.

B S1 is true and S2 is false.

C S1 is false and S2 is true.

D Both S1 and S2 are false.



[NAT]

#Q. Consider an augmented grammar

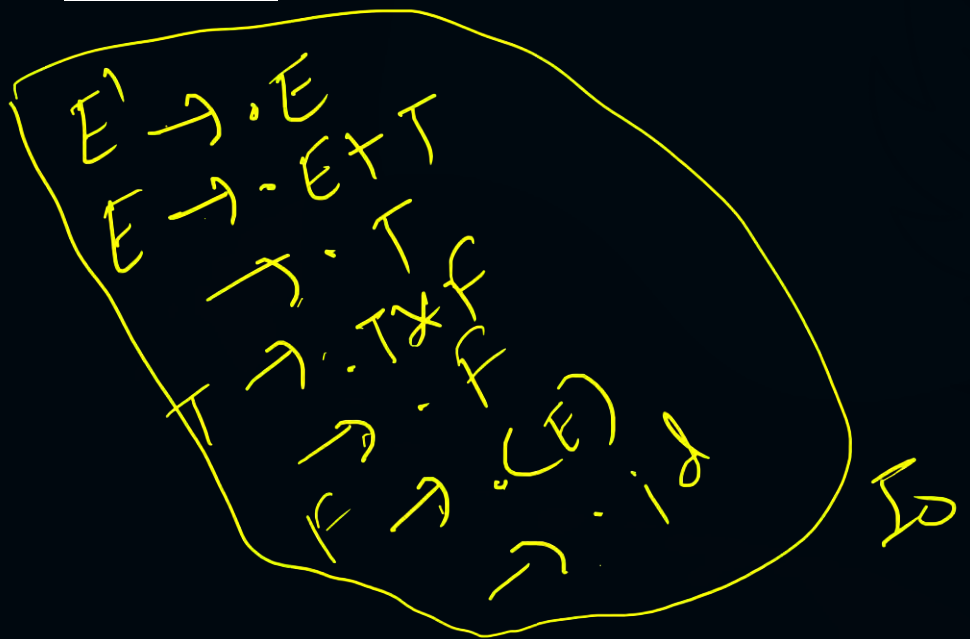
E'→E

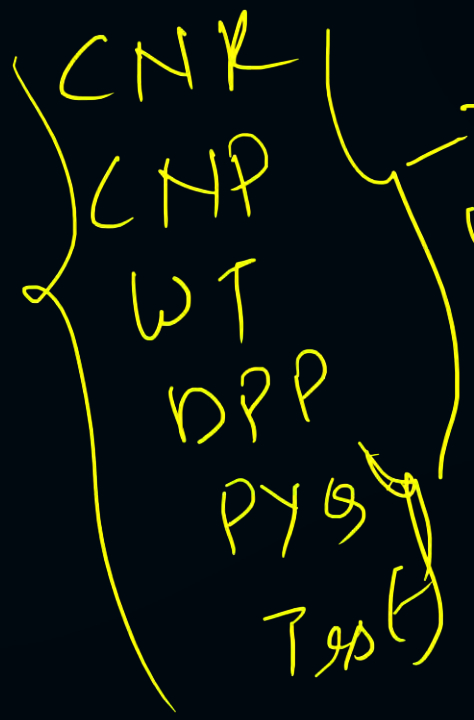
E→E+T | T

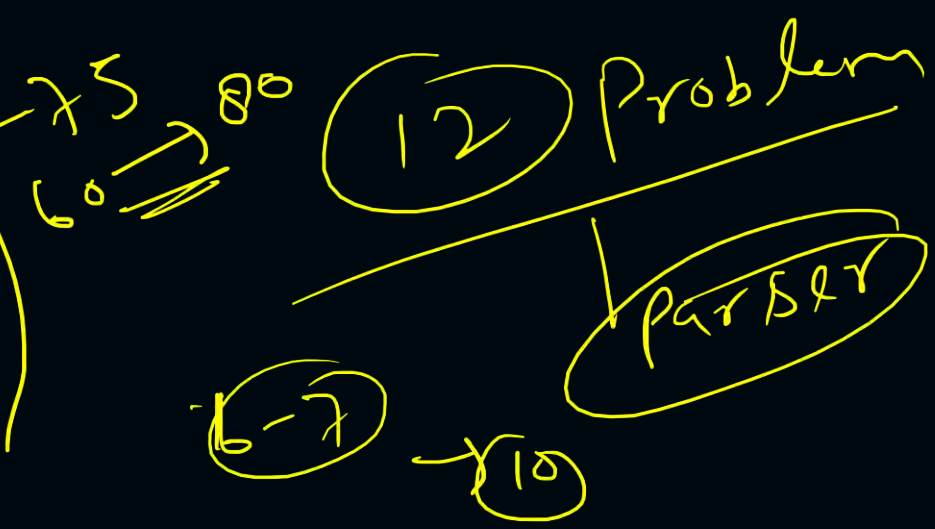
T→T*F | F

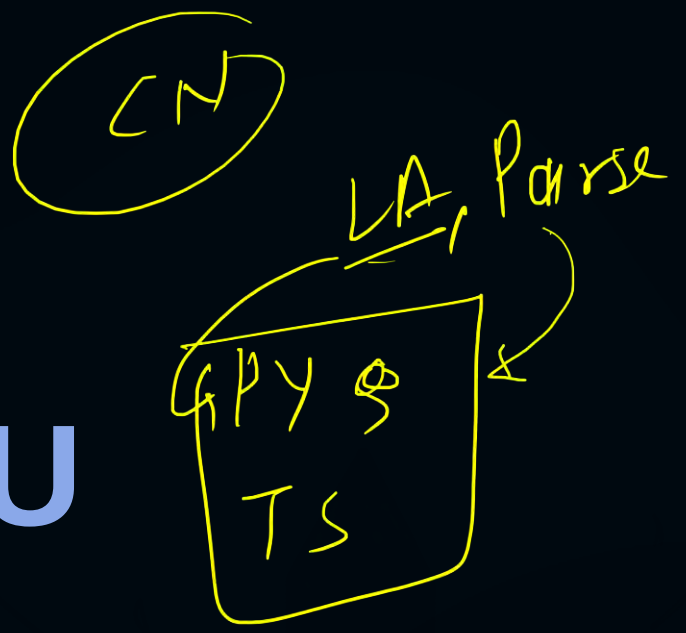
F→(E) | id

If I is the set of one item {[E'→.E]}, then number of items (including I) in CLOSURE(I) are _____




CNR
CNP
WT
DPP
PYQ
TYQ


75
60
80
12
Problem
Parser
6-7
10


CN
LA, Parse
PYQ
TS
Completion

THANK - YOU

